Eurowings Digital – Data Scientist

You can choose between one of the following two case studies:

Case Study 1:

Airfare Competitor Price Prediction – Take Home Test

Background

In the highly competitive airline industry, setting the right price for airfares is crucial for maximizing revenue and maintaining market share. Airfare pricing is not only about balancing demand and supply but also about strategic positioning in comparison to competitors. With the advent of dynamic pricing strategies, airlines now have the capability to adjust fares in real-time based on various factors including changes in demand, competitor prices, and other external variables.

By understanding the pricing landscape, airlines can better optimize their revenue management systems, enhancing their ability to predict future demand and adjust prices accordingly. This optimization directly impacts profitability and can be the difference between leading and lagging in the industry.

The Data: Using Skyscanner Data

For this task, you are required to use the Skyscanner data provided to you. Skyscanner is a leading global travel search site offering a comprehensive and free flight search service as well as instant comparisons for flights.

Data Description

You will be provided with data regarding user search behaviour and booking (redirect) patterns from Skyscanner aggregated in FlightWeeks. The data consist of:

- **Flight and passenger details:** Includes information about the FlightWeek, origin and destination airports and countries, the main airline carrier, and whether the flight is a connecting flight.
- **User details:** Information derived from the user's location and the type of trip.
- **Booking details:** Data on the cabin class, number of nights, booking horizon, number of redirects (bookings), and revenue generated. See table below for each column description:

Column	Description
'flightWeek'	The Calendar week where the first flight of the itinerary takes place
'OriginAirport'	
'DestinationAirport'	
'OriginCountry'	
'DestinationCountry'	
'mainCarrier'	The Airline with the longest flight distance in the itinerary searched (so the itinerary could have two operating airlines)
'isConnectingFlight'	

Column	Description
'isEWRoute'	Whether the route has EW operations on it
'Market'	Route (Origin-Destination combined)
'UserCountryCode'	IP location of user
'cabinClass'	
'kind'	The type of the itinerary: Return, One-way or multicity
'TripLengthNights'	Number of nights on the itinerary (if it's not one way)
'TravelHorizonDays'	Time period between redirect and first flight
'RedirectsCount'	Count of Bookings (redirects)
'Revenue'	Total value of itinerary (full booking)

So for the Pax numbers:	
An itinerary search for 3 people travelling LON-BER-DXB (connecting) and then returning DXB-LON (direct)	
Segments	3 flights x 3 people = 9 Segments
OD Pax	2 directions (LON-DXB return) x 3 people = 6 OD Pax
Trip Pax	3 people in redirect = 3 Trip Pax

Objective of the Take Home Test

The objective is to develop a predictive model using Skyscanner data that forecasts:

1. Future airfare prices of competitors.
2. Visualise the top influential features and provide reasoning.
3. (Bonus): Expected number of Eurowings (EW) redirects (indicative of bookings).
4. Visualise your solution through a visualization tool / product development.
5. Submit your solution in a Zip file (Python code that works, presentation slides).

Data

The dataset for this case study is provided as: ***skyscanner_airfare_data.csv***

Case Study 2:

Ancillary Pricing Prediction Case Study

Background

As part of our initiative to increase ancillary revenue per passenger, we want to optimize our ancillary prices. Currently, we optimize the prices of our flight-related ancillaries using machine learning models and analytical optimization.

Flight-related ancillaries are products such as seat reservations (standard seats, seats with more legroom), additional checked-in baggage, additional hand luggage, priority boarding, sports baggage, and more.

The following ancillaries are included in the given dataset:

Name	ProductName
First Baggage	1BA
Prio Boarding	BRD
Check In	CIP
Free Middle Seat	FMS
Seat Fee (Standard / More Legroom)	SEF STD (SDS) / SEF MLR
Pet in Cabin	PET
Second Baggage	2BA
Golf Sport Equipment	GOL
Bicycle	BIK
Board Equipment	SPQ

On our website and mobile app, users are split into two groups based on the pricing objective indicated by the column **"TreatmentOrdinal"**:

- **Control group (TreatmentOrdinal = 0):** Users see the standard price.
- **Treatment group (TreatmentOrdinal = 1):** Users see the optimized dynamic price.

We use the control group to calculate the benefit of the optimized model and to highlight if the model is producing more revenue than the standard price while maintaining or increasing the number of products bought.

Task

You have received a sample dataset containing randomized ancillary booking data. The "MaxExternalBookingID" is the hashed (anonymized) booking ID. The key of this table is the combination of:

- MaxExternalBookingID
- SegmentOriginLocationCode
- SegmentDestinationLocationCode

• BookingUTC

For every flight on every booking, you will find one row for each product with the information about ancillary pricing. The "QuantityBooked" column denotes how many of the specified product were bought for the given key. The AmountReport illustrates the price shown to the customer for the specific product. The OriginalPrice is the standard price shown to the control group (TreatmentOrdinal = 0).

Even if no product was booked, the products are listed with a QuantityBooked value of "0". This allows us to compare if products are being bought and at what prices (for both the control group and the treatment group). A segment refers to one passenger flying from A to B. A connecting flight like A-B-C for one passenger is counted as two segments (A to B and B to C).

Tasks / Questions

1. **Exploratory Data Analysis (EDA):** Perform EDA to understand the data distribution, trends, and patterns. What insights can you gather about the ancillary products and their bookings?
2. **Model Building:** Build a machine learning model to optimize the pricing of the ancillary products. Describe your modeling approach, feature selection, and the algorithms used.
3. **Performance Measurement:** Define KPIs to measure the performance of the pricing model. How would you determine which group (control or treatment) performs better for each product?
4. **Product and Route Analysis:** Provide insights into which products and routes perform best and worst in the treatment group compared to the control group.
5. **Dashboarding:** Put your analysis into a dashboarding tool (PowerBI preferred) to visualize the results.

Bonus

1. Analyze the data with Python to derive additional insights.
2. Provide suggestions on what other information or details could help in analyzing ancillary dynamic pricing.

Dataset

The dataset for this case study is provided as: ***ancillary_pricing_data.csv***

Submission Instructions

We have designed the assignment to be completed within a week. If you think it will take longer, please let us know so we can work something out. Since our mail filter rejects external emails with zip attachments, we ask that you upload the solution to a host of your choice (e.g. Github, Dropbox, Google Drive). Please make sure that the solution is **not** publicly available.

If you have any questions, please don't hesitate to contact us.